School of Future Tech

Case Study Report

on

Smart Traffic Signal Simulation

by

Jui Sudhir Tawde

Roll No: 150096725149

Index

1. Introduction to the Case Study.
2. Problem Statement / Case Background (Abstract).
3. Problem Statement / Case Study Design.
4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study.
5. Problem Statement / Case Study Implementation Details and Snapshots.
6. Problem Statement / Case Study Results and Conclusion.
7. References

1. Introduction to the Case Study

Traffic signals are critical components in modern urban traffic management systems. They regulate the flow of vehicles at intersections to minimize congestion and enhance safety. Traditional traffic signal systems follow fixed time intervals, which can lead to inefficient traffic flow under varying conditions.

This case study focuses on developing a **Smart Traffic Signal Simulation** using C++. The system simulates the sequential operation of traffic lights (Green, Yellow, Red) with predefined time intervals. It demonstrates how sequential execution, time-based logic, and control system concepts can be implemented in a programming environment to simulate real-world systems.

The simulation provides a visual interface using **ANSI color codes** and **emojis**, displays countdown timers for each signal, and allows the user to start, stop, or exit the signal sequence interactively.

---

2. Problem Statement / Case Background (Abstract)

Background

Managing traffic efficiently requires well-timed signal systems. Many intersections use static timers, which do not account for real-time traffic conditions. Understanding and simulating signal timing is essential for designing efficient traffic control systems.

Manually analyzing signal sequences and timing logic can be cumbersome. Therefore, a **programmatic simulation** that accurately cycles through signals with visual indicators and countdown timers is useful for both educational and practical purposes.

Abstract

This project presents the design and implementation of a **Smart Traffic Signal Simulation** using C++. The system:

- Simulates three signal states: Green, Yellow, Red
- Uses time-based countdowns for each signal state

- Displays the current state with colors and emoji indicators
- Allows the user to start, stop, or exit the simulation
- Demonstrates sequential execution and control system logic

The project highlights concepts of **time-based programming, sequential logic, and real-world system simulation** in a C++ console application.

---

3. Case Study Design

The Smart Traffic Signal Simulation is designed as a **console-based C++ program** with a menu-driven interface.

1. Signal States

The system implements three signals:

- **Green** 🟢 – Signals vehicles to move
- **Yellow** 🟡 – Indicates caution, preparing to stop
- **Red** 🔴 – Signals vehicles to stop

Each signal is displayed with a corresponding **ANSI color code** for visual differentiation.

---

2. Signal Timing Logic

The system follows predefined durations for each signal state:

Signal	Duration (Seconds)
Green	30
Yellow	5
Red	30

A **countdown timer** is displayed for each state to simulate the remaining time before the next signal change. This demonstrates **time-based execution**.

---

3. Program Flow

The simulation follows a **menu-driven program flow**:

1. Initialize the system
2. Display the main menu:
    - Start Signal
    - Stop Signal
    - Exit
3. On **Start Signal**, the system cycles through Green → Yellow → Red repeatedly
4. After completing a cycle, the user can choose to **Continue** or **Stop**
5. On **Stop Signal**, the simulation stops after the current cycle
6. Exit terminates the program

The flow ensures both **control logic** and **interactive simulation**.

---

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key Methods Used:

<u>1. Sequential Execution</u>

The program cycles through signal states using a **while loop**, ensuring the correct **sequence of signals** is maintained: Green → Yellow → Red.

<u>2. Time-Based Simulation</u>

The **countdown function** implements delays using `this_thread::sleep_for()` in C++:

- Simulates real-time passage for each signal state
- Updates the console every second to show the remaining time

3. Interactive User Menu

The system includes a **menu-driven interface**:

- User chooses to start, stop, or exit the simulation
- After each signal cycle, the user is prompted to **Continue or Stop**
- ANSI colors and emojis provide a visual indication of the current signal

4. Console-Based Visualization

- Signals are displayed with **color codes** for clarity
- Countdown is printed in the console
- Emojis indicate the current signal visually

Technology Stack Used

Programming Language

- **C++** – Console-based simulation, sequential logic, and time-based control

Libraries / Features

- `<iostream>` – Input/output handling
- `<thread>` and `<chrono>` – Time-based delays (`sleep_for`)

Design Techniques

- Menu-driven console interface
- ANSI color coding for visual feedback
- Emojis for signal representation

5. Problem Statement / Case Study Implementation Details and Snapshots.

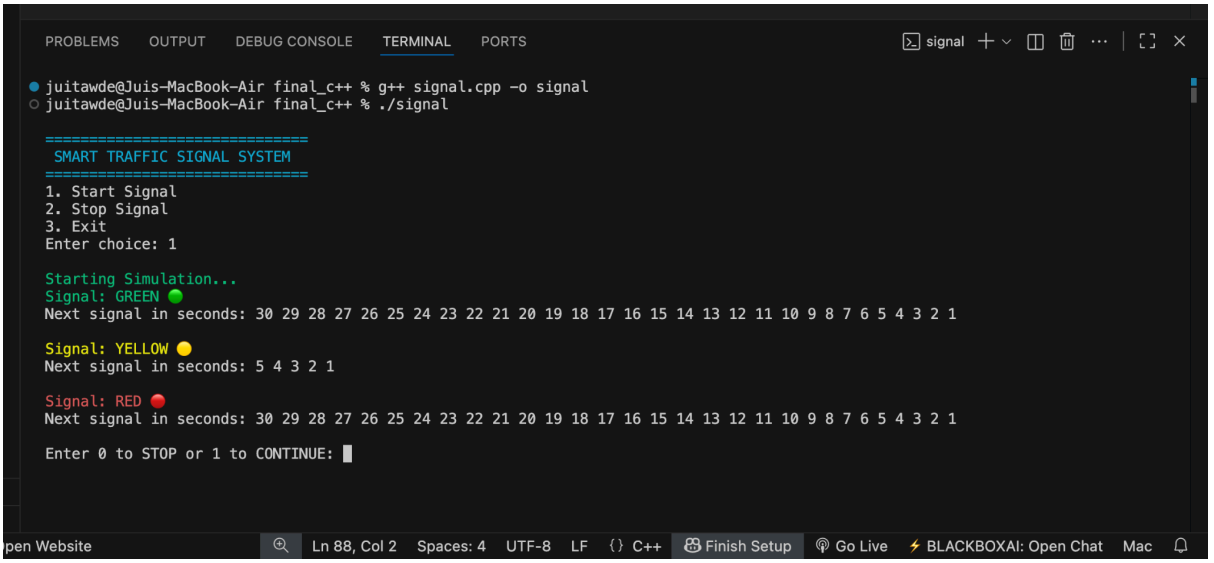
Implementation Structure

The project consists of a single **C++ source file** implementing the following modules:

1. **Main Menu** – Presents Start, Stop, Exit options
2. **Signal Cycle** – Cycles through Green, Yellow, Red signals sequentially
3. **Countdown Function** – Displays remaining seconds for each signal
4. **User Interaction** – Allows stopping or continuing simulation after each cycle

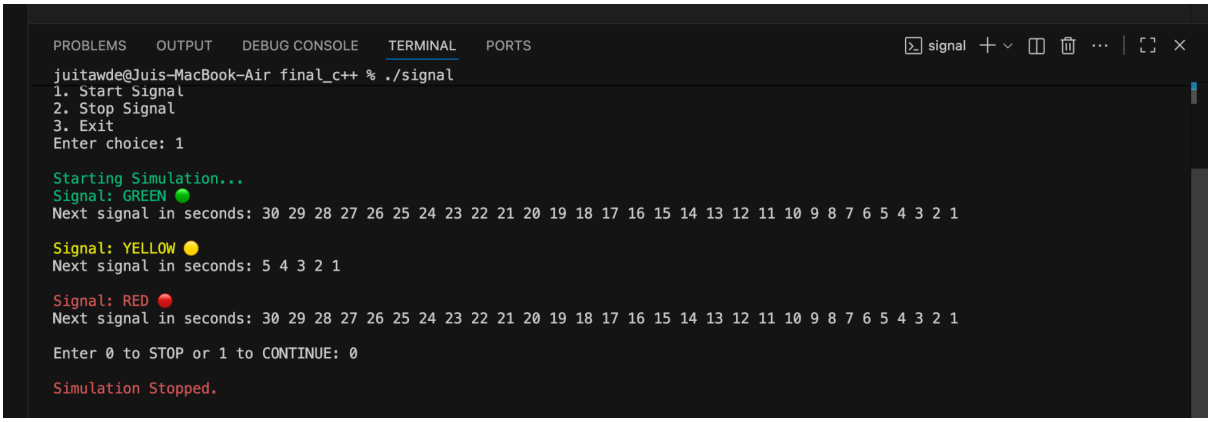
Snapshots you can include:

- Console displaying **Signal countdown**



- Main menu interface showing Start/Stop/Exit

- Simulation stopped message after user stops



- Exiting the program after user chooses exit

---

6. Problem Statement / Case Study Results and Conclusion.

Findings

The project successfully demonstrates:

- Correct **signal sequencing**: Green → Yellow → Red
- Proper **time-based simulation** with countdowns
- Interactive **menu-driven control**
- Use of **ANSI color codes** and emojis for visual clarity

Conclusion

The **Smart Traffic Signal Simulation** project demonstrates the application of C++ programming to real-world control systems. The simulation:

- Implements **three signal states** with visual indicators
- Uses **time-based logic** to change signals sequentially
- Provides an **interactive menu** for user control
- Enhances understanding of **control system logic** and **time-based simulations**

The project can be extended in the future to include:

- Real-time traffic condition adjustment for dynamic signal timing
- Graphical interface for improved visualization
- Multi-intersection simulation.

7. References

- C++ Reference – Input/Output Streams
  https://en.cppreference.com/w/cpp/io
- C++ Reference – Thread Library
  https://en.cppreference.com/w/cpp/thread
- ANSI Escape Codes Documentation
  https://en.wikipedia.org/wiki/ANSI_escape_code